TECHNISCHE UNIVERSITÄT DARMSTADT

Tutorial "Machine Learning in Solid Mechanics"

Task 3: Viscoelasticity
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Cyber-Physical Simulation
TU Darmstadt

Outline

Viscoelastic Maxwell model

Recurrent neural networks

Organisation

Coding

1D linear elasticity

Consider a 1D bar with stress $\sigma \in \mathbb{R}$ and strain $\varepsilon \in \mathbb{R}$. Similar to 3D hyperelasticity, we can introduce the internal energy e as

$$e(\varepsilon) = \frac{1}{2} E \, \varepsilon^2 \,, \tag{1}$$

where E is the Young's modulus. The corresponding stress follows as

$$\sigma = \frac{\partial e(\varepsilon)}{\partial \varepsilon} = E \, \varepsilon \,. \tag{2}$$

This constitutive model is thermodynamic consistent since the Clausius-Duhem inequality

$$\sigma \dot{\varepsilon} - \dot{e} \geq 0 \tag{3}$$

is fulfilled, where $\dot{a} = \frac{\partial a}{\partial t}$ denotes the time derivative of a.

Task: Show that Eq. (1) is thermodynamic consistent.

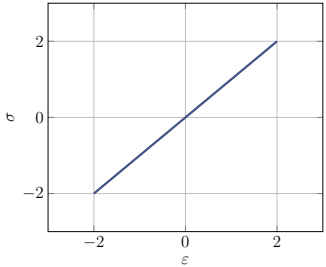
Solution:

$$\begin{aligned} \sigma \dot{\varepsilon} - \dot{e} &= \sigma \dot{\varepsilon} - \frac{\partial e(\varepsilon)}{\partial \varepsilon} \dot{\varepsilon} \\ &= \sigma \dot{\varepsilon} - E \, \varepsilon \, \dot{\varepsilon} \\ &= (\sigma - E \, \varepsilon) \, \dot{\varepsilon} = 0 \,. \end{aligned} \tag{4}$$

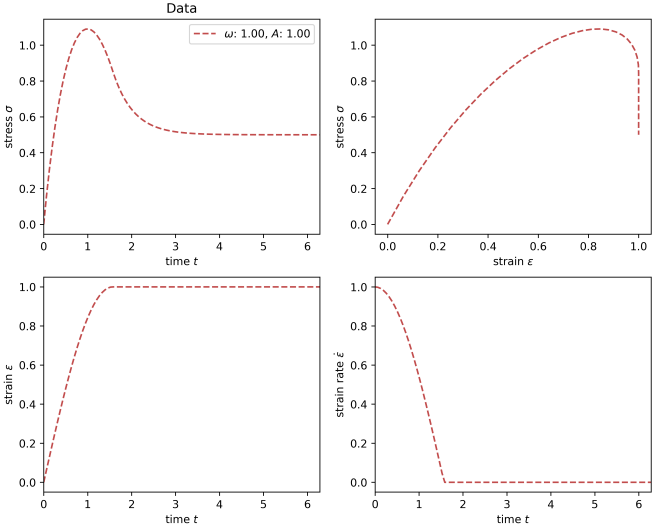
In the elastic case, thermodynamic consistency is fulfilled when the stress is defined as the gradient of the internal energy.

1D linear elasticity

What stress relationship can be represented by a linear elastic model?



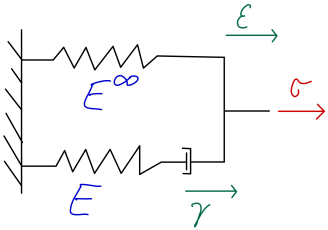
$$e(\varepsilon) = \frac{1}{2} E \, \varepsilon^2 \,, \qquad \sigma = E \, \varepsilon \tag{5}$$



Viscoelastic Maxwell model

- The Maxwell model [1]consists of an equilibrium spring with stiffness $E^\infty > 0$ in parallel to a series connection of a non-equilibrium spring with stiffness $E > 0$ and a damper with viscosity $\eta > 0$.
- The internal variable γ denotes the strain acting on the damper. Consequently, the strain $(\varepsilon - \gamma)$ acts on the non-equilibrium spring.
- Energy is stored in both springs, leading to the internal energy e and the stress σ

$$e(\varepsilon, \gamma) = \frac{1}{2} E^\infty \varepsilon^2 + \frac{1}{2} E (\varepsilon - \gamma)^2 \,, \qquad \sigma = E^\infty \varepsilon + E (\varepsilon - \gamma) \,. \tag{6}$$



- With the damper behaving like a Newtonian fluid

$$\sigma^{\text{damper}} = \eta \, \dot{\gamma} \,, \tag{7}$$

and by setting $\sigma^{\text{damper}} = \sigma^{\text{non-equilibrium}}$, the **evolution equation** for the internal variable follows as

$$\dot{\gamma} = \frac{E}{\eta} (\varepsilon - \gamma) \,. \tag{8}$$

[1] G. A. Holzapfel. *Nonlinear Solid Mechanics: A Continuum Approach for Engineering*. 2nd ed. Wiley, 2000. ISBN: 9780471823193, Chapter 6.9.

Thermodynamic consistency of the Maxwell model

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Task: Show that the Maxwell model

$$
\begin{aligned}
e(\varepsilon,\,\gamma) &= \frac{1}{2}E^{\infty}\,\varepsilon^2 + \frac{1}{2}E\,(\varepsilon-\gamma)^2\,,\\
\sigma &= E^{\infty}\,\varepsilon + E\,(\varepsilon-\gamma)\,,\\
\dot{\gamma} &= \frac{E}{\eta}(\varepsilon-\gamma)\,,
\end{aligned}
\tag{9}
$$

is thermodynamic consistent by inserting it in the Clausius-Duhem inequality

$$\sigma\dot{\varepsilon} - \dot{e} \geq 0\,. \tag{10}$$

Solution:

$$
\begin{aligned}
&\sigma\,\dot{\varepsilon} - \frac{\partial e(\varepsilon,\,\gamma)}{\partial\varepsilon}\dot{\varepsilon} - \frac{\partial e(\varepsilon,\,\gamma)}{\partial\gamma}\dot{\gamma} \geq 0\\
&\qquad\qquad\qquad\qquad \Leftrightarrow\\
&\left(\sigma - \frac{\partial e(\varepsilon,\,\gamma)}{\partial\varepsilon}\right)\dot{\varepsilon} - \frac{\partial e(\varepsilon,\,\gamma)}{\partial\gamma}\dot{\gamma} \geq 0\,.
\end{aligned}
\tag{11}
$$

With Eq. $(22)_2$, the left term of above equation vanishes. The second term is the **dissipation inequality**

$$\mathcal{D} = -\frac{\partial e(\varepsilon,\,\gamma)}{\partial\gamma}\dot{\gamma} = \frac{E^2}{\eta}(\varepsilon-\gamma)^2 \geq 0\,. \tag{12}$$

Calculation of γ

- The strain ε (=input) and the stress σ (=output) are *measurable / observable* variables.
- In contrast to that, γ is an *internal variable*. This variable can't be directly measured, but follows from the evolution equation and an initial condition:

$$t_0 = 0: \qquad \varepsilon_0 = \sigma_0 = \gamma_0 = 0\,, \qquad \dot{\gamma} = \frac{E}{\eta}(\varepsilon - \gamma)\,. \tag{13}$$

- Thus, the calibration / test datasets have the form

$$\mathcal{D} = \{\Delta t_1,\, \varepsilon_1,\, \sigma_1;\, \ldots,\, \Delta t_m,\, \varepsilon_m,\, \sigma_m;\}\,, \tag{14}$$

with $\Delta t_n = t_{n+1} - t_n$.

- How can we calculate $\gamma(t)$ for arbitrary complex deformation histories $\varepsilon(t)$?

- Explicit Euler scheme:

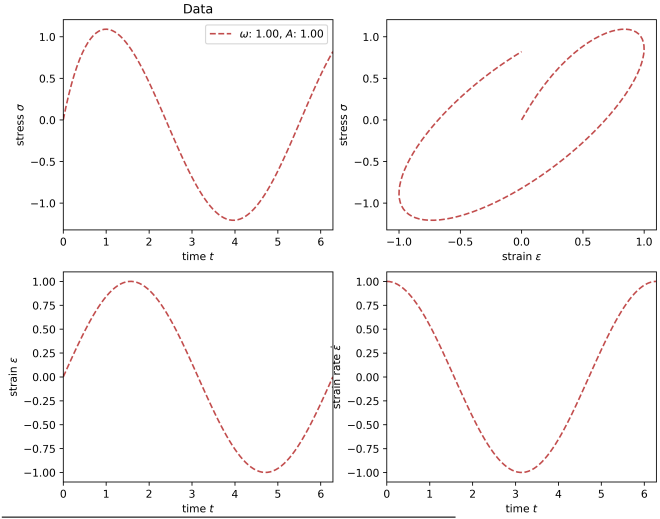
$$\begin{aligned} \gamma_{n+1} &= \gamma_n + \Delta t_n\, \dot{\gamma}_n \\ &= \gamma_n + \Delta t_n\, \frac{E}{\eta}\,(\varepsilon_n - \gamma_n)\,. \end{aligned} \tag{15}$$

- Example:

$$\begin{aligned} \gamma_0 &= 0\,, \\ \gamma_1 &= \gamma_0 + \Delta t_0 \frac{E}{\eta}\,(\varepsilon_0 - \gamma_0) = 0\,, \\ \gamma_2 &= \gamma_1 + \Delta t_1 \frac{E}{\eta}\,(\varepsilon_1 - \gamma_1) = \Delta t_1 \frac{E}{\eta}\,\varepsilon_1\,, \\ \gamma_3 &= \gamma_2 + \Delta t_2 \frac{E}{\eta}\,(\varepsilon_2 - \gamma_2)\,, \\ &\vdots \end{aligned} \tag{16}$$

- How should we chose Δt_n?

Harmonic oscillation



- Parameters:[2]

$$\{E^\infty, E, \eta\} = \{0.5, 2, 1\}\,. \tag{17}$$

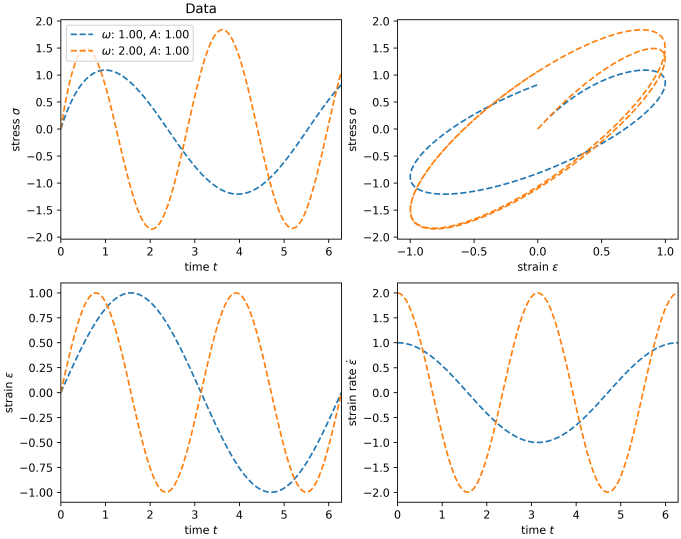
- Strain and strain rate:

$$\begin{aligned} \varepsilon(t) &= A\,\sin(\omega\,t)\,, \\ \dot{\varepsilon}(t) &= A\,\omega\,\cos(\omega\,t)\,, \\ t &\in [0,\,2\pi]\,. \end{aligned} \tag{18}$$

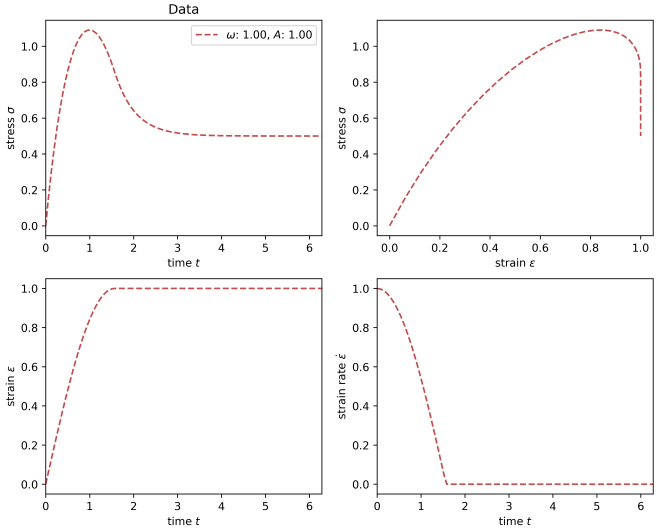
- 100 discrete points in time

[2] C.-Y. Lin et al. "Constitutive equations for analyzing stress relaxation and creep of viscoelastic materials based on standard linear solid model derived with finite loading rate". In: *Polymers* 14.10 (2022). DOI: 10.3390/polym14102124, Table 1.

Harmonic oscillation - variation of the strain rate



Relaxation



- Apply the strain as in Eq. (18) until the first maximum of ε and keep it fixed afterwards

Generalisations of the Maxwell model

Consider the constitutive equations of the standard Maxwell model

Standard Maxwell model

$$
\begin{aligned}
e(\varepsilon, \gamma) &= \frac{1}{2} E^{\infty} \varepsilon^2 + \frac{1}{2} E (\varepsilon - \gamma)^2 \,, \\
\sigma &= E^{\infty} \varepsilon + E (\varepsilon - \gamma) \,, \\
\dot{\gamma} &= \frac{E}{\eta} (\varepsilon - \gamma) \,.
\end{aligned}
\tag{19}
$$

How can we generalise this model to increase its flexibility?

- Use a nonlinear evolution equation
- Use a nonlinear energy potential

General evolution equation for the internal variable

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At first, we stick to the internal energy and the stress

$$e(\varepsilon,\,\gamma) = \frac{1}{2}E^{\infty}\,\varepsilon^2 + \frac{1}{2}E\,(\varepsilon-\gamma)^2\,,$$
$$\sigma = E^{\infty}\,\varepsilon + E\,(\varepsilon-\gamma)\,, \tag{20}$$

and make a nonlinear ansatz for the evolution equation

$$\dot{\gamma} = f(\varepsilon,\,\gamma)(\varepsilon-\gamma), \qquad f(\varepsilon,\gamma) > 0\,. \tag{21}$$

Task: Show that Eq. (21) is a thermodynamic consistent choice of evolution equation.

Solution:

$$\sigma\,\dot{\varepsilon} - \frac{\partial e(\varepsilon,\,\gamma)}{\partial\varepsilon}\dot{\varepsilon} - \frac{\partial e(\varepsilon,\,\gamma)}{\partial\gamma}\dot{\gamma} \geq 0$$
$$\Leftrightarrow \tag{22}$$
$$\left(\sigma - \frac{\partial e(\varepsilon,\,\gamma)}{\partial\varepsilon}\right)\dot{\varepsilon} - \frac{\partial e(\varepsilon,\,\gamma)}{\partial\gamma}\dot{\gamma} \geq 0\,.$$

Again, with the definition of σ, the left term of above equation vanishes with. The second term is the **dissipation inequality**

$$\mathcal{D} = -\frac{\partial e(\varepsilon,\,\gamma)}{\partial\gamma}\dot{\gamma} = f(\varepsilon,\,\gamma)(\varepsilon-\gamma)^2 E \geq 0\,. \tag{23}$$

Generalized standard materials (GSM)

In the GSM framework[3], in addition to the energy potential $e(\varepsilon, \gamma)$, a dissipation potential $\phi = \phi(\varepsilon, \gamma, \dot{\gamma})$ is introduced. Here, we use

$$\phi(\varepsilon, \gamma, \dot{\gamma}) = \frac{1}{2g(\varepsilon, \gamma)}\dot{\gamma}^2, \quad g(\varepsilon, \gamma) > 0\,, \tag{24}$$

and let e remain a general function. The stress σ is given as

$$\sigma = \frac{\partial e(\varepsilon, \gamma)}{\partial \varepsilon}\,. \tag{25}$$

For GSM models, the additional constitutive relation

$$\frac{\partial e(\varepsilon, \gamma)}{\partial \gamma} + \frac{\partial \phi(\varepsilon, \gamma, \dot{\gamma})}{\partial \dot{\gamma}} = 0\,, \tag{26}$$

is introduced.

Task: Use Eqs. (26) and (24) to derive $\dot{\gamma}$ and show the thermodynamic consistency of the model.

Solution: Inserting Eq. (24) in Eq.(26) yields

$$\begin{aligned} &\frac{\partial e(\varepsilon, \gamma)}{\partial \gamma} + \frac{1}{g(\varepsilon, \gamma)}\dot{\gamma} = 0 \\ &\qquad\qquad\qquad\qquad \Leftrightarrow \\ &\dot{\gamma} = -g(\varepsilon, \gamma)\frac{\partial e(\varepsilon, \gamma)}{\partial \gamma}\,. \end{aligned} \tag{27}$$

Inserting Eq. (27) in the dissipation inequality yields

$$\mathcal{D} = -\frac{\partial e(\varepsilon, \gamma)}{\partial \gamma}\dot{\gamma} = g(\varepsilon, \gamma)\left(\frac{\partial e(\varepsilon, \gamma)}{\partial \gamma}\right)^2 \geq 0\,. \tag{28}$$

[3] A. Kumar and O. Lopez-Pamies. "On the two-potential constitutive modeling of rubber viscoelastic materials". In: *Comptes Rendus Mécanique* 344.2 (2016), pp. 102–112. ISSN: 1631-0721. DOI: 10.1016/j.crme.2015.11.004.

Overview of the models

How can the models be transitioned to the standard Maxwell model?

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General evolution equation

$$
\begin{aligned}
e(\varepsilon,\,\gamma) &= \frac{1}{2}E^{\infty}\,\varepsilon^2 + \frac{1}{2}E\,(\varepsilon-\gamma)^2\,,\\
\sigma &= E^{\infty}\,\varepsilon + E\,(\varepsilon-\gamma)\,,\\
\dot{\gamma} &= f(\varepsilon,\,\gamma)(\varepsilon-\gamma), \qquad f(\varepsilon,\gamma) > 0\,.
\end{aligned}
\tag{29}
$$

Standard Maxwell model

$$
\begin{aligned}
e(\varepsilon,\,\gamma) &= \frac{1}{2}E^{\infty}\,\varepsilon^2 + \frac{1}{2}E\,(\varepsilon-\gamma)^2\,,\\
\sigma &= E^{\infty}\,\varepsilon + E\,(\varepsilon-\gamma)\,,\\
\dot{\gamma} &= \frac{E}{\eta}(\varepsilon-\gamma)\,.
\end{aligned}
\tag{31}
$$

GSM model

$$
\begin{aligned}
e &= e(\varepsilon,\,\gamma)\\
\sigma &= \frac{\partial e(\varepsilon,\,\gamma)}{\partial\varepsilon}\,.\\
\dot{\gamma} &= -g(\varepsilon,\gamma)\frac{\partial e(\varepsilon,\gamma)}{\partial\gamma}\,, \quad g(\varepsilon,\gamma) > 0\,.
\end{aligned}
\tag{30}
$$

- The evolution equation $\dot{\gamma}$ can be solved with an explicit Euler scheme for each model
- The approach based on a general evolution equation can be transitioned to the standard Maxwell model by setting $f(\varepsilon,\,\gamma) = \frac{E}{\eta} = \text{const.}$
- The GSM model can be transitioned to the standard Maxwell model by setting Eq. $(30)_1$ as Eq. $(31)_1$ and setting $g := \eta^{-1} = \text{const.}$

Feed-forward neural networks (FFNNs)

- Consider the (single-layered) FFNN with input $\boldsymbol{x} \in \mathbb{R}^l$ and output $\boldsymbol{y} \in \mathbb{R}^m$:

$$\boldsymbol{y} = \boldsymbol{w}^2 \, \mathcal{SP}\left(\boldsymbol{w}^1 \boldsymbol{x} + \boldsymbol{b}^1\right) + \boldsymbol{b}^2 \,. \tag{32}$$

- Consider the time series

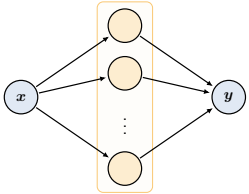
$$\mathcal{D} = \{\boldsymbol{x}_1(t_1),\, \boldsymbol{y}_1(t_1);\, \ldots\, \boldsymbol{x}_n(t_n),\, \boldsymbol{y}_n(t_n)\} \,. \tag{33}$$

  Can we apply FFNNs for time series?

- Yes, as *direct solution models*:[4]

$$\boldsymbol{y}_k = \mathsf{FFNN}(\boldsymbol{x}_k,\, t_k) \,. \tag{34}$$

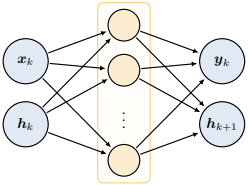
- However, this leads to a poor model generalisation.

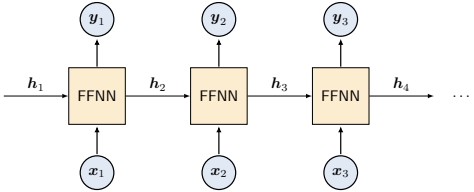


[4] Qinyu Zhuang et al. "Model order reduction based on Runge–Kutta neural networks". en. In: *Data-Centric Engineering* 2 (2021), e13. ISSN: 2632-6736. DOI: 10.1017/dce.2021.15.

Recurrent neural networks (RNNs)

- RNNs[5] allow to propagate information throughout a time series
- This is done by introducing an internal / state variable $\boldsymbol{h}$
- Basically, a FFNN is evaluated repeatedly over time





[5] S. Kollmannsberger et al. *Deep Learning in Computational Mechanics*. Vol. 977. Studies in Computational Intelligence. Springer, 2021. ISBN: 978-3-030-76586-6. DOI: 10.1007/978-3-030-76587-3, Chapter 3.10.2.

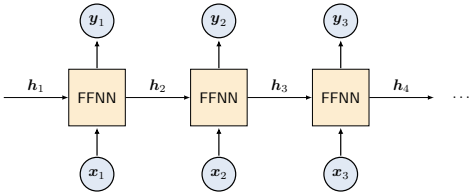
How can we adapt RNNs for the Maxwell model?

General evolution equation

$$
\begin{aligned}
e(\varepsilon,\,\gamma) &= \frac{1}{2}E^{\infty}\,\varepsilon^2 + \frac{1}{2}E\,(\varepsilon-\gamma)^2\,,\\
\sigma &= E^{\infty}\,\varepsilon + E\,(\varepsilon-\gamma)\,,\\
\dot{\gamma} &= f(\varepsilon,\,\gamma)(\varepsilon-\gamma), \qquad f(\varepsilon,\gamma) > 0\,.
\end{aligned}
\tag{35}
$$

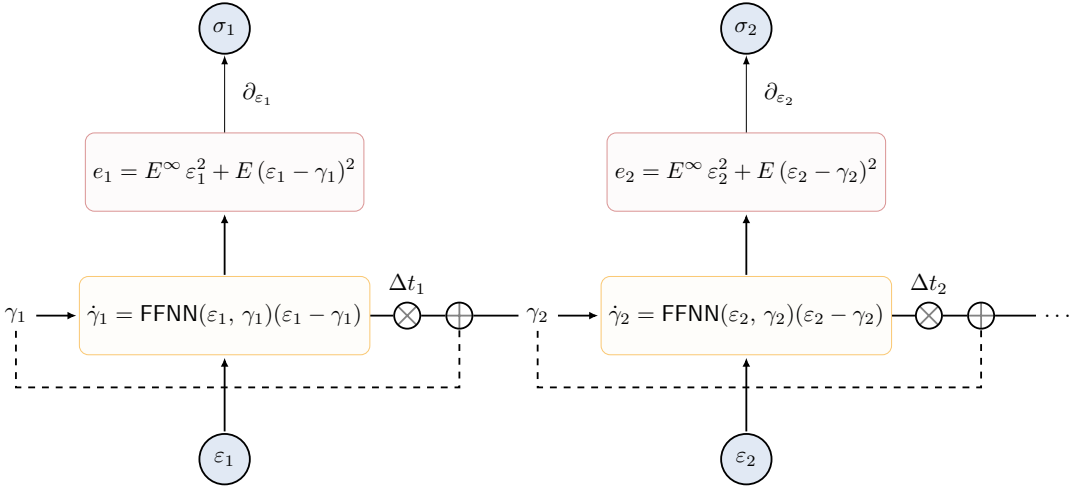
- We can represent f by a FFNN, i.e., $f(\varepsilon,\,\gamma) = \mathsf{FFNN}(\varepsilon,\,\gamma)$.
- How do we ensure $f > 0$?
- This yields the explicit Euler scheme:

$$
\begin{aligned}
\gamma_{n+1} &= \gamma_n + \Delta t_n\,\dot{\gamma}_n\\
&= \gamma_n + \Delta t_n\,\mathsf{FFNN}(\varepsilon_n,\,\gamma_n)\,(\varepsilon_n-\gamma_n)\,.
\end{aligned}
\tag{36}
$$



Maxwell model with general evolution equation – NN model

This is a special type of Runge-Kutta neural network (RKNN)



Organisation

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- 09. + 10.02.2026: Result presentation task 3.
  - Presentation length: 10 minutes.
  - Afterwards: Grading of the last presentation and final grade of the tutorial.
  - Feedback round.
- Now: Implementation of the standard Maxwell model in KLAX based on the code provided on GitHub.

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